

Tyler Stoeger

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EDUCATION

University of Illinois Chicago

Ph.D. in Economics

M.A. in Applied Economics

CMEP Exchange Scholar, University of Chicago Harris School of Public Policy

Ph.D. coursework: Energy and Environmental Economics III (Spring); I (Fall)

Chicago, IL

Expected May 2027

May 2024

2024

Purdue University Northwest

Bachelor of Science in Business, Major in Finance

Minors in Economics and Applied Mathematics

GPA: 4.00/4.00; Highest Distinction

Hammond, IN

May 2022

RESEARCH FIELDS

Environmental and Resource Economics; Applied Microeconomics; Public Economics

JOB MARKET PAPER

Behavioral Responses to Air Quality Information: Implications for Optimal Policy

2026 Graduate Silvia Saffrin Award for Excellence in the Third-Year Paper

Abstract. Air quality alerts can improve welfare by helping individuals avoid pollution exposure, but frequent warnings may reduce responsiveness and weaken the effectiveness of the alert system. This paper studies how people's outdoor activity responds to air quality alerts, how that response varies with recent alert history, and how these behavioral dynamics affect the optimal alert threshold. A model of optimal alert policy is combined with a fuzzy regression discontinuity design using air quality forecasts, alert issuances, and high-frequency foot traffic data from six U.S. metropolitan areas. Near the alert threshold, receiving an alert reduces daily visits to outdoor recreational points of interest by approximately 3.3 visits, or 9 percent relative to the sample mean. The response varies substantially with recent alert history: alerts reduce visits by roughly eight visits when one or two alerts have occurred during the previous seven days, but have little detectable effect after three or more recent alerts. A counterfactual simulation calibrated to New York City-specific pollution dynamics and behavioral responses places the welfare-maximizing alert threshold at approximately 80, below the current AQI threshold of 101. The results show that optimal warning policy depends not only on the pollution risks communicated by an alert, but also on how the frequency and sequencing of repeated warnings affect behavioral responsiveness.

WORKING PAPERS

Air Quality Forecasts in Practice

Draft expected Oct. 2026

Documents systematic differences between official air-quality forecasts and realized AQI and investigates the statistical and institutional mechanisms underlying those differences. The analysis uses a broad panel of AirNow forecasts and realized air quality across more years and geographic areas than the job market paper, with planned analyses of forecast-error distributions, distributional differences, and serial dependence.

REFERENCES

Matthew Tarduno (Dissertation Chair)

Assistant Professor of Economics, University of Illinois Chicago

tarduno@uic.edu

Benjamin Feigenberg

Professor of Economics, University of Illinois Chicago

bfeigenb@uic.edu

Ben Ost

Professor of Economics, University of Illinois Chicago

bost@uic.edu

HONORS & AWARDS

Graduate Silvia Saffrin Award, University of Illinois Chicago

2026

Awarded for excellence in the third-year paper

Chancellor's Medallion, Purdue University Northwest

2022

Awarded to graduates earning the highest GPA in their academic college

RESEARCH GRANTS

Award for Graduate Research, UIC Graduate College	2025
\$2,900 in research support	
Undergraduate Research Grant, Purdue University Northwest	Fall 2021
Undergraduate Research Grant, Purdue University Northwest	Spring 2021

PRESENTATIONS

2026	
Southern Economic Association Annual Meeting, Graduate Student Session, Houston, TX	Forthcoming
2025	
University of Chicago, Environmental and Energy Economics Graduate Student Workshop	December
University of Illinois Chicago, Economics Active Research Lunch (EARL)	November
2024	
University of Illinois Chicago, Economics Active Research Lunch (EARL)	November

RESEARCH EXPERIENCE

University of Illinois Chicago	Fall 2024; Summer 2025
Research Assistant to Matthew Tarduno	
Conducted literature reviews, collected and cleaned data, performed preliminary empirical analysis, and edited manuscripts for two research projects.	
Purdue University Northwest	Jan. 2021 – May 2022
Research Assistant to Amlan Mitra	
Conducted literature reviews, collected and cleaned data, and supported preliminary empirical analysis across faculty and funded undergraduate research projects.	

TEACHING EXPERIENCE

University of Illinois Chicago	
Instructor of Record	
ECON 300 - Econometrics	Summer 2024; Spring 2025; Fall 2025
Designed and taught lectures, assignments, exams, and applied statistical-programming labs; managed a teaching assistant and held office hours.	
Teaching Assistant	
ECON 300 - Econometrics	Summer 2023; Spring 2024; Fall 2026
ECON 534 - Econometrics I	Fall 2023; Fall 2024
ECON 121 - Principles of Macroeconomics	Fall 2022; Spring 2026
ECON 120 - Principles of Microeconomics	Spring 2023; Fall 2023
ECON 220 - Microeconomics: Theory and Applications	Spring 2023
ECON 510 - Microeconomic Theory II	Spring 2024
Graduate Assistant	
ECON 516 - Development Economics	Spring 2026

ADDITIONAL TRAINING

Berkeley/Sloan Summer School in Environmental and Energy Economics, UC Berkeley	2025
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PROFESSIONAL MEMBERSHIPS

Association of Environmental and Resource Economists (AERE); American Economic Association (AEA); Southern Economic Association (SEA)

TECHNICAL SKILLS

Software: R, Stata, MATLAB, LaTeX, Git/GitHub, Excel.
Additional: Spatial data analysis in R, including distance calculations, point-in-polygon operations, and geographic data processing.